



Semester One Examination, 2024

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One: Calculator-free

SOLUTIONS

WA student number: In figures

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In words

Circle your Teacher's Name: Mr Bartoshefski Mrs Fraser-Jones Ms Horley Dr Kang
Ms Lastari Ms Nathan Mr Singh

Time allowed for this section

Reading time before commencing work: five minutes

Working time: fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener,
correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	7	7	50	52	35
Section Two: Calculator-assumed	10	10	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free**35% (52 Marks)**

This section has **seven** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(4 marks)

- (a) Determine the value of the expression $\frac{15}{x^2 - \sqrt{y}}$ when $x = 4$ and $y = 36$. (2 marks)

Solution
$\frac{15}{4^2 - \sqrt{36}} = \frac{15}{16 - 6} = \frac{15}{10} = \frac{3}{2} = 1.5$
Specific behaviours
✓ simplifies denominator ✓ simplifies fraction or obtains decimal equivalent

- (b) Use the formula $A = \frac{h}{2}(a + b)$ to determine the value of A when $a = 15$, $b = 27$ and $h = 10$. (2 marks)

Solution
$A = \frac{10}{2}(15 + 27) = \frac{10}{2} \times 42 = 10 \times 21 = 210$
Specific behaviours
✓ correctly substitutes and simplifies bracketed term ✓ correct value of A

Question 2

(8 marks)

- (a) An item is priced at \$135 excluding GST. Determine the price of the item once 10% GST has been included. (2 marks)

Solution
$135 + 13.50 = \$148.50$
Specific behaviours
✓ correctly indicates 10% of price ✓ correct price including GST

- (b) A homeowner is offered a 2% discount for early payment of their \$1700 annual council rates bill. How much would they pay if they took up the offer? (2 marks)

Solution
$1\% \rightarrow \$17, \quad 2\% \rightarrow \34 $1700 - 34 = \$1666$
Specific behaviours
✓ correctly indicates 2% of bill ✓ correct amount

- (c) A watch enthusiast paid \$500 for a timepiece but later sold it for just \$440. Determine the percentage loss made by the enthusiast. (2 marks)

Solution
$500 - 440 = 60$ $\frac{60}{500} = \frac{6}{50} = \frac{12}{100} = 12\% \text{ loss}$
Specific behaviours
✓ correctly indicates loss ✓ correct percentage loss, must state 'loss' or '-12%'

- (d) A gallery charges a 35% commission on the price of all artworks that it sells. If it sells a small sculpture for \$560 what commission will it charge? (2 marks)

Solution
$10\% \rightarrow \$56, \quad 5\% \rightarrow \$28, \quad 20\% \rightarrow \112 Commission is $56 + 28 + 112 = \$196$.
Specific behaviours
✓ indicates any correct method to obtain commission ✓ correct commission

Question 3

(10 marks)

Use these four matrices to answer the following questions.

$$\mathbf{A} = \begin{bmatrix} 5 & -2 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 2 & 6 \\ 5 & -4 \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} 4 & 6 \\ 5 & -2 \end{bmatrix} \quad \mathbf{D} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}$$

- (a) Two of the following matrix calculations are
- not**
- possible.

$$\mathbf{AB} \quad \mathbf{BC} \quad \mathbf{BA} \quad \mathbf{B} + \mathbf{C} \quad \mathbf{C} + \mathbf{D}$$

Circle the two that are not possible and explain why.

(4 marks)

Solution	
BA is not possible as the number of rows in B is not the same as the number of rows in A .	
C+D is not possible as they do not have the same dimensions.	
Specific behaviours	
✓ Circles BA .	
✓ Correct explanation of why BA cannot be determined. Accept: 2×2 <u>1</u> $\times 2$ <i>not the same</i>	
✓ Circles C + D .	
✓ Correct explanation of why C + D cannot be determined.	

- (b) Determine
- $\mathbf{C} - 2\mathbf{B}$
- .

(2 marks)

Solution	
$\begin{bmatrix} 4 & 6 \\ 5 & -2 \end{bmatrix} - 2 \begin{bmatrix} 2 & 6 \\ 5 & -4 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 5 & -2 \end{bmatrix} - \begin{bmatrix} 4 & 12 \\ 10 & -8 \end{bmatrix}$	
$= \begin{bmatrix} 0 & -6 \\ -5 & 6 \end{bmatrix}$	
Specific behaviours	
✓ correct scalar multiple	
✓ simplifies correctly to a single matrix	

- (c) Determine
- $\mathbf{CB}$
- .

(2 marks)

Solution	
$\mathbf{CB} = \begin{bmatrix} 4 & 6 \\ 5 & -2 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ 5 & -4 \end{bmatrix} = \begin{bmatrix} 38 & 0 \\ 0 & 38 \end{bmatrix}$	
Specific behaviours	
✓ any 2×2 matrix	
✓ correct product	

- (d) Determine a matrix
- $\mathbf{P}$
- which, when added to
- $\mathbf{B}$
- , results in a zero matrix.

(2 marks)

Solution	
$\mathbf{P} + \begin{bmatrix} 2 & 6 \\ 5 & -4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$	
$\mathbf{P} = \begin{bmatrix} -2 & -6 \\ -5 & 4 \end{bmatrix}$	
Specific behaviours	
✓ Writes down a suitable equation or recognises that $\mathbf{P}$ is the opposite of $\mathbf{B}$.	
✓ Determines $\mathbf{P}$.	

Question 4

(8 marks)

- (a) A free veterinary clinic at a show treated 11 skin, 8 stomach and 5 other ailments on Saturday and the following day treated 7 skin, 6 stomach and 9 other ailments. Express the numbers for Saturday in row matrix A , do the same for the Sunday numbers in row matrix B , and then determine matrix $A + B$ that represents the numbers of each type of ailment treated over the weekend. (2 marks)

Solution
$A + B = [11 \ 8 \ 5] + [7 \ 6 \ 9] = [18 \ 14 \ 14]$
Specific behaviours
✓ correct row matrices (any consistent order of elements) ✓ writes and correctly obtains sum of matrices

- (b) A stall at the same event on Saturday sold small show bags for \$15 each and large show bags for \$20 each. The sales for this stall are shown in the table below.

	Small	Large
First hour	6	11
Second Hour	10	16

- (i) Represent the number of bags sold in the 2×2 matrix N . (1 mark)

Solution
$N = \begin{bmatrix} 6 & 11 \\ 10 & 16 \end{bmatrix}$
Specific behaviours
✓ correct square matrix

- (ii) Represent the costs of a small and a large show bag in matrix C so that the matrix product NC will produce a meaningful result. (1 mark)

Solution
$C = \begin{bmatrix} 15 \\ 20 \end{bmatrix}$
Specific behaviours
✓ correct column matrix

- (iii) Determine the matrix $X = NC$, and explain what the entry $x_{1,1}$ represents. (3 marks)

Solution
$X = \begin{bmatrix} 6 & 11 \\ 10 & 16 \end{bmatrix} \begin{bmatrix} 15 \\ 20 \end{bmatrix} = \begin{bmatrix} 310 \\ 470 \end{bmatrix}$ $x_{1,1}$ represents the stall's income from the sale of show bags in the first hour.
Specific behaviours
✓ one correct entry in matrix X ✓ correct matrix X ✓ states income from sale of show bags in the first hour

- (iv) Determine the stall's total income from the sale of show bags during the first two hours of the event. (1 mark)

Solution
Total income is $310 + 470 = \$780$.
Specific behaviours
✓ correct income, with dollar sign

Question 5

(7 marks)

Information about two companies listed on a stock exchange is shown in the table below.

Company Code	Share price (\$)	Annual earnings per share (cents)	Last dividend paid per share (cents)
CBQ	1.44	3	5
QLM	2.00	5	7

- (a) Determine the last dividend payment to an investor who owned 4200 shares in CBQ.

(2 marks)

Solution
$4200 \times 5 \div 100 = 42 \times 5 = \210 or $4200 \times 5 = 21\,000 \text{ cents}$
Specific behaviours
✓ indicates correct expression for payment ✓ correct dividend

- (b) The last dividend paid per share by QLM was 7 cents when its share price was \$2.00. Determine the equivalent percentage dividend.

(2 marks)

Solution
$\frac{7}{200} = \frac{3.5}{100} \therefore 3.5\%$
Specific behaviours
✓ indicates correct expression for percentage ✓ correct percentage

Solution
$\frac{0.07}{2} \times 100 \therefore 3.5\%$
Specific behaviours
✓ indicates correct expression for percentage ✓ correct percentage

- (c) Calculate the price-to-earnings ratio for each company.

(2 marks)

Solution
CBQ: $\frac{1.44}{3} = 48$, QLM: $\frac{2.00}{5} = 40$ or CBQ: $\frac{1.44}{0.03} = 48$, QLM: $\frac{2}{0.05} = 40$
Specific behaviours
✓ correct ratio for CBQ ✓ correct ratio for QLM

- (d) State, with justification, in which of the two companies an investor would pay the least for each dollar of the company's earnings.

(1 mark)

Solution
QLM – it has the lowest P/E ratio.
Specific behaviours
✓ correct company, with justification that references P/E ratio

Question 6

(8 marks)

- (a) A floor plan of a home is drawn where 4 cm on the plan represents 2.4 m in the actual home. The scale factor for this plan is $1:n$, where n is a whole number. Determine the value of n . (2 marks)

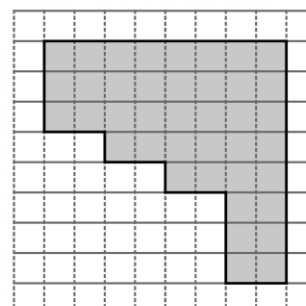
Solution
2.4 m is 240 cm.
Hence ratio is 4: 240 which simplifies to 1: 60.
Hence $n = 60$.
Specific behaviours
✓ writes a correct ratio after eliminating units
✓ correct value of n

- (b) The side lengths of triangle A are 5 cm, 7 cm and 8 cm. The side lengths of triangle B are 20 cm, 28 cm and 36 cm. Are these triangles similar figures? Justify your answer. (3 marks)

Solution
Ratio of shortest sides: $20 \div 5 = 4$. Ratio of middle sides: $28 \div 7 = 4$. Ratio of longest sides: $36 \div 8 = 4.5$.
Hence the triangles are not similar as the ratios of corresponding sides are not the same for all three pairs of sides.
Specific behaviours
✓ correctly calculates two ratios of side lengths that show not equal ✓ states triangles are not similar ✓ correctly justifies why triangles are not similar

Solution
$\frac{20}{5} \neq \frac{36}{8}$ or $\frac{28}{7} \neq \frac{36}{8}$
Hence the triangles are not similar as the ratios of corresponding sides are not the same for all three pairs of sides.
Specific behaviours
✓ correctly compares two ratios of side lengths that show not equal. ✓ states triangles are not similar ✓ correctly justifies why triangles are not similar

- (c) The shaded region on the grid to the right is a scale drawing of a concrete slab, where each small square has a side length of 4 mm. If the area of the actual concrete slab is 360 square metres, determine the scale factor of the drawing. (3 marks)



Solution
There are 40 shaded squares and so the area of one square is $360 \div 40 = 9 \text{ m}^2$.
Hence side of one square must be $\sqrt{9} = 3 \text{ m}$.
Hence scale factor is $4: 3000 = 1: 750$.
Specific behaviours
✓ indicates correct area of one square ✓ correct side length of one square ✓ correct scale factor, simplified

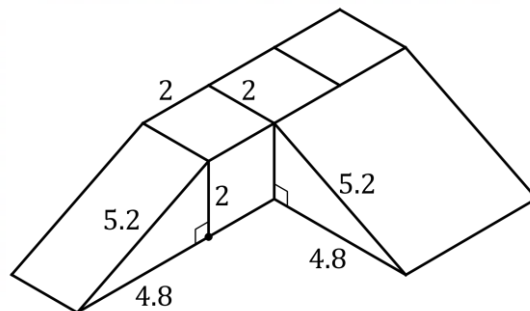
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Question 7

(7 marks)

A large three-dimensional sculpture is to be suspended outside a sports arena to advertise a skateboarding competition. The sculpture comprises three two-metre cubes and two triangular prisms that will be glued together to form the composite figure as shown in the diagram (not to scale).

All dimensions are in metres and the triangular cross-section of each prism has sides of 5.2 m, 4.8 m and 2 m.



- (a) Show that the volume of the smaller triangular prism is 9.6 cubic metres.

(1 mark)

Solution
Area of cross-section: $A = \frac{1}{2}bh = \frac{1}{2} \times 4.8 \times 2 = 4.8 \text{ m}^2$
Volume of prism: $V = Ah = 4.8 \times 2 = 9.6 \text{ m}^3$
Specific behaviours
✓ shows correct expression or combination of expressions

- (b) Determine the volume of the composite figure.

(2 marks)

Solution
Volume of one cube is $2 \times 2 \times 2 = 8 \text{ m}^3$. Total volume is sum of three cubes and three small triangular prisms: $V = 3 \times 8 + 3 \times 9.6$ $= 24 + 28.8$ $= 52.8 \text{ m}^3$
Specific behaviours
✓ indicates volume of one cube ✓ correct total volume

Solution
$V = 2^3 \times 3 + 9.6 + 2 \times \frac{1}{2} \times 4.8 \times 2 \times 2$ $= 24 + 9.6 + 19.2$ $= 52.8 \text{ m}^3$
Specific behaviours
✓ indicates volume of one cube ✓ correct total volume

- (c) The sculpture needs to be coated with a waterproof sealant spray. Determine the total surface area of the composite figure that needs to be covered. Show clear evidence of the method you use.

(4 marks)

Solution
There are 11 exposed faces of the cubes: $A_1 = 11 \times 2 \times 2 = 44$
There are 4 triangular faces: $A_2 = 4 \times 4.8 = 19.2$
There are 4 rectangular faces of the triangular prisms: $A_3 = 2 \times 5.2 + 4 \times 5.2 + 2 \times 4.8 + 4 \times 4.8 = 60$
Hence the total surface area is: $\text{TSA} = 44 + 19.2 + 60 = 123.2 \text{ m}^2$
Specific behaviours
✓ clearly shows method employed ✓ obtains or indicates correct method for one of A_1, A_2 or A_3 ✓ obtains a second A_1, A_2 or A_3 ✓ correct total surface area, including units

End of questions

Supplementary page

Question number: _____

